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Apple Recruitment Team

Bengaluru, Karnataka, India

Subject: Application for Service Reliability Engineer - Apple Data Platform

Dear Hiring Manager,

I am writing to express my strong interest in the Service Reliability Engineer position within the Apple Data Platform (ADP Processing SRE) team. With a deep background in managing high-throughput data infrastructure and a passion for reducing operational toil through Python-based automation, I am highly aligned with Apple's mission to deliver secure, reliable, and performant data analytics services at a global scale.

My familiarity with Apple's engineering standards began during my time as a Network & DevOps Trainee supporting an Apple Infrastructure Project at Tata Consultancy Services (TCS). In that role, I developed a strong foundation in Linux environments and networking fundamentals by troubleshooting deep OS issues using strace, tcpdump, and Wireshark. I hardened server security, deployed Nginx load balancers, and configured strict network firewalls to resolve complex routing failures.

Currently, as a Data Engineer at TCS managing enterprise-scale infrastructure, I operate directly on the front lines of production reliability. I am responsible for ensuring the availability and performance of a high-throughput, event-driven AWS architecture (utilizing Apache Kafka/MSK, EventBridge, and Lambda) that processes 200 GB to 500 GB of IoT data daily. My day-to-day involves managing automated CI/CD deployment lifecycles using GitHub Actions and Docker, ensuring that continuous delivery pipelines are robust and secure.

A core part of my engineering philosophy aligns with your goal of building self-healing systems. Recently, I engineered a Python and Boto3-based automation script that completely replaced manual validation checklists. This tool is now adopted by our entire 30-person DevOps team to proactively monitor system workflows. By deeply integrating Datadog and AWS CloudWatch, I built alerting mechanisms for low-latency applications that successfully reduced manual operational toil by 90%. Furthermore, I continuously drive Root Cause Analysis (RCA) on Docker container vulnerabilities, patching critical CVEs to harden the architecture and minimize downtime.

My degree in Mechanical Engineering fundamentally shaped my systems-thinking mindset, training me to view complex platforms as interconnected ecosystems where observability is paramount. Recently, I published academic research through Jadavpur University Press (ISBN: 978-81-993635-8-8) focusing on data-driven mathematical modeling and algorithmic optimization using meta-heuristic algorithms. I apply this exact analytical rigor to capacity forecasting, performance tuning, and optimizing distributed data pipelines.

I am fully prepared to operate within your hybrid work model in Bengaluru and would welcome the opportunity to discuss how my proactive approach to incident management can contribute to the ADP Processing SRE team.

Thank you for your time and consideration.

Sincerely,

Akashdip Mahapatra